

Mohammad Zaid

GenAI Engineer & Architect · 5+ Years · TCS ♀ Lucknow, India

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Nationality: Indian | Visa: Eligible for Work Visa | Notice: 15 Days | Open to Relocation: GCC/GULF



PROFESSIONAL SUMMARY

GenAI Architect & AI Engineer with 5+ years at Tata Consultancy Services (TCS), specializing in **production-grade LLM systems, agentic pipelines, and RAG architectures** for enterprise clients across Healthcare and BFSI. Hands-on with LangChain, LangGraph, Claude AI, FastAPI, and cloud AI platforms (Azure OpenAI, AWS Bedrock). Delivered measurable outcomes - 60% documentation overhead reduction, 80%+ design-system compliance, 70% faster PR review cycles. Experienced team lead (4–6 engineers) with regulatory depth across FCA, MAR, DFSA, and CBUAE frameworks. Open to Senior GenAI Architect / Agentic AI Engineering Lead roles across the GCC region.

CORE COMPETENCIES

LLM Orchestration & Prompt Engineering · RAG Pipeline Design · Agentic AI Workflows · LangChain / LangGraph / LangSmith · Claude AI / GPT-5 / Gemini · CrewAI / AutoGen · FastAPI · MLOps / LLMOps · Vector Databases (FAISS, Chroma, Pinecone) · Azure AI & AWS Bedrock · Figma-to-React Automation · AST-based Code Analysis · KDB+/q Time-Series Analytics · Arabic NLP Awareness · Team Leadership (4–6 engineers) · FCA / MAR / DFSA / CBUAE Compliance

EXPERIENCE

Tata Consultancy Services · System Engineer (GenAI Architect & Team Lead)

New Delhi | Apr 2021 – Present

◆ Healthcare - GenAI Architect & Team Lead (Python · LangChain · LangGraph · Claude AI · Agentic AI)

- Architected an AI application that auto generates business logic from code flow diagrams using LangChain, LLMs, and Python cutting manual analysis time by ~50% across engineering teams.
- Designed production RAG pipelines with vector stores (FAISS) and LLMs for context aware document retrieval, improving knowledge extraction accuracy over large enterprise codebases.
- Built a legacy Java code reverse engineering solution using Claude AI + LangChain + RAG + custom AST based parser to auto generate structured technical documentation reducing documentation overhead by ~60% and accelerating modernization.
- Developed an end to end Figma to React pipeline: Claude AI for semantic design interpretation, Playwright for DOM traversal and visual validation, and internal MUI component library achieving 80%+ design system compliance and eliminating design engineering handoff cycles.
- Designed and built a multi agent PR review system on LangGraph that automatically analyses pull requests on GitHub parsing diffs, identifying logic errors, flagging security anti patterns, and posting structured inline review comments via the GitHub API.

◆ BFSI - Capital Markets Trade Surveillance (KDB+/q · SDL · SDP)

- Optimised a real-time trade surveillance system monitoring FICC trades for misconduct detection under FCA and MAR regulatory frameworks, improving alert precision across multi asset classes.
- Maintained SDL/SDP pipelines in KDB+/q for high throughput, low latency tick data ingestion; authored high-performance time series queries and led root cause analysis for production data integrity issues.

◆ Aviation - Database Management & Optimisation (SQL · Linux)

- Managed and optimised relational databases for aviation operations; built SQL based ETL pipelines with indexing and partitioning strategies powering analytics ready dashboards.

Dweepi Infotech · Intern - Systems & Data

New Delhi, India | Jul 2019 – Dec 2019

- Worked on Linux kernel internals including device driver development, interrupt handling, and memory management.
- Assisted in code review and testing of kernel patches, collaborating with senior engineers on system-level debugging and documentation.

KEY SKILLS

Generative AI & LLMs: LangChain, LangGraph, LangSmith, LlamaIndex, CrewAI, AutoGen, OpenAI API, Anthropic Claude API, Prompt Engineering, RAG, Agentic Workflows, Fine-tuning, MCP Servers

Vector Databases: FAISS, Chroma, Pinecone, OpenAI Embeddings, text-embedding models

Python & AI Tooling: Python, FastAPI, Hugging Face Transformers, Playwright, AST parsing, API integration & automation

MLOps / LLMOps: LangSmith, Docker, Docker Compose, CI/CD, Git, Kafka, Redis Streams

Cloud AI: Azure AI Services / OpenAI on Azure, AWS Bedrock, Vercel

Data & Databases: PostgreSQL, TimescaleDB, Redis, KDB+/q, SQL, ETL pipelines, time-series analytics

Agile & Leadership: Team Lead (4–6 engineers), Jira, Rally, Sprint Planning, cross-functional deliver

CERTIFICATIONS

- LangChain for LLM Application Development
- Building Systems with the ChatGPT API
- Generative AI with Large Language Models - AWS
- Microsoft Certified: Azure AI Fundamentals
- KX Advanced Course (Level 1 / 2 / 3) - KX Systems
- Data Analysis with Python - freeCodeCamp

EDUCATION

Dr APJ Abdul Kalam Technical University · B. Tech - Computer Science & Engineering

Lucknow | Aug 2015 – Jun 2019

SSV Inter College · Intermediate - Mathematics, Physics, Chemistry

Ayodhya | Jul 2013 – May 2015

SSV Inter College · High School - Science

Ayodhya | Jul 2012 – May 2013

PERSONAL PROJECTS

FileForge(ForgeTech) - Browser-Native AI File Toolkit | Next.js · TypeScript · ONNX Runtime Web · TensorFlow.js · Tesseract.js · FFmpeg · Tailwind

🔗 <https://forgetech.vercel.app> | GitHub: github.com/mohd98zaid/FileForge

- Built a browser-native platform with 185+ file tools (image, PDF, audio, video, developer utilities) that processes all files entirely client-side using WebAssembly and ONNX - zero file uploads, zero backend, zero privacy risk.
- Integrated AI-powered background removal via ONNX Runtime Web, OCR across 100+ languages via Tesseract.js, and audio/video conversion via FFmpeg WASM - all running in the browser without any server roundtrip.
- Built for Indian users with full Hindi/English bilingual UI, SSC/UPSC/NEET/Railway exam photo presets, and passport/visa photo tools - serving both domestic and international audiences.
- Deployed on Vercel with auto-generated sitemap, SEO metadata per tool page, and PWA support.

RepairDeskz - AI-Powered Repair Shop Management System | Next.js · FastAPI · PostgreSQL · Redis · WhatsApp Business API · Three.js · GSAP · R3F

🔗 <https://repairdeskz.vercel.app> | GitHub: github.com/mohd98zaid/RepairDesk

- Full-stack PWA for independent repair shops covering the complete ticket lifecycle - drop-off, diagnostics, parts tracking, invoicing, digital signature capture, and customer pickup notification.
- Integrated WhatsApp Business API for automated customer notifications on every ticket status change; integrated an AI assistant for natural-language shop analytics ("Which technician closed most tickets this week?").
- Built real-time inventory management with auto-deduction on ticket completion, low-stock alerts, supplier management, and per-part margin tracking.
- Cinematic 3D landing page built with React Three Fiber (R3F), GSAP scroll-driven animations, and Lenis inertial scrolling - showcasing enterprise-grade frontend engineering.

QuantDSS - Quantitative Trading Decision Support System | FastAPI · Python · Next.js ·

PostgreSQL/TimescaleDB · Redis · Kafka · Docker

🔗 <https://github.com/mohd98zaid/QuantDSS>

- Self-hosted intraday trading platform for NSE/BSE equities with live market data ingestion via Upstox/Angel One WebSockets, 11-layer signal intelligence pipeline, and a 17-rule risk engine with hard circuit breakers.
- Strategies include EMA Crossover, RSI Mean Reversion, ORB+VWAP, Volume Expansion, and VWAP Reclaim - each signal passes deduplication, meta-strategy blocking, ML/NLP filtering, and liquidity checks before reaching the risk engine.
- Chaos-tested to enterprise standard: distributed locking for concurrent signal safety, PEL recovery workers, session-isolated broker retries, zombie position handling, and auto-recovery from Redis exhaustion and broker failures.
- Supports Paper Trading and Live Execution modes with Kafka + Redis Streams dual-write pipeline, Telegram bot notifications, Prometheus/Grafana observability, and full tick data lake for backtesting.